1 Definition and Examples

It is a general principle in mathematics that when one defines an interesting mathematical structure, one should also study the "structure-preserving" maps (functions) between two such structures. In the case of vector spaces, the fundamental operations are addition and scalar multiplication, or, equivalently, forming linear combinations. A structure-preserving map between vector spaces is thereofre one which preserves linear combinations .These are called linear maps.

Definition 1.1 (Linear map)

Let V and W be vector spaces over F. A *linear map* is a function $T : V \to W$ with the property that

$$T(\alpha v_1 + v_2) = T(v_1) + T(v_2). \tag{1}$$

The terms *linear function* or *linear transformation* are often used interchangeably with *linear map*. These all have exactly the same meaning. If $W = V$, then a linear map $T : V \to V$ is said to be a *linear operator* on V.

If T is a linear map and v is a vector, it is common to write Tv instead of $T(v)$. These two notations also have exactly the same meaning.

Exercise 1.2

Shwo that the condition in Equation (1) is equivalent to the two conditions

$$T(v_1 + v - 2) = Tv_1 + Tv_2$$
$$T(\alpha v) = \alpha Tv$$

for all $v, v_1, v_2 \in V$ and $\alpha \in F$. Some books use these two conditions instead of Equation (1). Other books define a linear map to be one which satisfies

$$T(\alpha_1 v_1 + \alpha_2 v_2) = \alpha_1 Tv_1 + \alpha_2 Tv_2$$

for all $v_1, v_2 \in V$ and all $\alpha_1, \alpha_2 \in F$. It is easily verified that this condition is also equivalent to the other two.

Solution 1

First, we show (1.2) implies this. By choosing $\alpha = 1$ and arbitrary v_1, v_2, we recover the first condition. By choosing $v_2 = 0$ and arbitrary $v_1.v_2$, we recover the second condition. Conversely, applying the first condition to $alphav_1, v_2 \in V$ (a small abuse of notation) and v_2 be v_2, by the first condition we see $T(\alpha v_1 + v_2) = T(\alpha v_1) + T(v_2)$, and applying the second condition to the first term gives $T(\alpha v_1 + v_2) = \alpha T(v_1) + T(v_2)$.

Example 1.3

Take $W = V$ and consider the identity map $\mathrm{id}_V : V \to V$, defined by $\mathrm{id}_V(v) = v$ for every $v \in V$. In $v_1, v_2 \in V$ and $\alpha \in F$, then

$$\begin{aligned}\mathrm{id}_V(\alpha v_1 + v_2) &= \alpha v_1 + v_2 \\ &= \alpha \mathrm{id}_V v_1 + \mathrm{id}_V v_2,\end{aligned}$$

which shows that id_V is linear.

Exercise 1.4

Let $0 : V \to W$ denote the zero map, defined by $0v = 0$ for all $v \in V$. Show that the *zero map*, defined by $0v = 0$ for all $v \in V$. Show that the zero map is linear. (Note that 0 on the left-hand side of this equation denotes the zero map, while 0 on the right-hand side denotes the zero vector

of the vector space W. It is common to use the same notation with different meanings when the context makes clear what each use is supposed to mean.)

Proof:

We have $0(\alpha v_1 + v_2) = 0 = 0 + 0 = \alpha \cdot 0 + 0 = \alpha 0(v_1) + 0(v_2)$, showing linearity as desired.

Example 1.5

Let $\mathcal{P}(\mathbf{R})$ denote the set of all polynomials with real coefficients. Since the derivative of another polynomial is another polynomial, we can view differentiation as a map $D : \mathcal{P}(\mathbf{R}) \to \mathcal{P}(\mathbf{R})$. Two of the basic properties of differentiation say that $(f + g)' = f' + g'$ and $(\alpha f)' = \alpha f'$ showing that D is a linear operator on $\mathcal{P}(\mathbf{R})$

Exercise 1.6

Define $T : \mathcal{P}(\mathbf{R} \to \mathbf{R}$ by

$$Tp = \int_0^1 p(x)\,\mathrm{d}x\,.$$

Show that T is a linear map.

Proof:

We know that (bounded) integrals are linear by the axioms for an integral (in the BC-Fast style, this can be proved rigorously via either a Darboux or Riemann approach).

Exercise 1.7

Define $T : \mathcal{P}(\mathbf{R}) \to \mathcal{P}(\mathbf{R})$ by $(Tp)(x) = x^2 p(x)$ for each $x \in \mathbf{R}$. Show that T is a linear map.

Proof:

In fact, multiplication of polynomials by $\cdot_g(f) = f \cdot g$ is always a linear map (the most general statement is that in the algebra of polynomials, $\cdot$ is a bilinear map $\mathcal{P}(\mathbf{R}) \times \mathcal{P}(\mathbf{R}) \to \mathcal{P}(\mathbf{R})$). For, we have:

$$\begin{aligned}
\cdot_g(\alpha f + h)(x) &= g(x)((\alpha f + h))(x) \\
&= g(x)(\alpha f(x) + h(x)) \\
&= g(x)\alpha f(x) + g(x)h(x) \\
&= \alpha(g(x)f(x)) + g(x)h(x) \\
&= \alpha \cdot_g (f(x)) + \cdot_g(h(x)),
\end{aligned}$$

showing linearity as desired.

Exercise 1.8

Let F^ω denote the vector space of all sequences of elements of F. Define $T : F^\omega \to F^\omega$ by $T(x_1, x_2, x_3, ...) = (x_2, x_3, ...)$. Show that T is a linear operator on F^ω.

Proof:

Let $(x_i), (y_i)$ be sequences. $\alpha(x_i) + (y_i)$ is the sequence $(\alpha x_1, +y_1, \alpha x_2 + y_2, ...)$ Then, $T(\alpha(x_i) + (y_i)) = T(\alpha x_1, +y_1, \alpha x_2 + y_2, ...) = (\alpha x_2 + y_2, ...) = \alpha(x_i)' + (y_i)' = \alpha T(x_i) + T(y_i)$, where $(x_i)'$ and $(y_i)'$ are the sequences $(x_i)$with the first term removed.

Exercise 1.9

Define $T : \mathbf{R}^3 \to \mathbf{R}^2$ by $T(x, y, z) = (2x - y + 3z, 7 + 5y - 6z)$. Prove that T is a linear map.

Proof:

This is automatic from the fact that the components of $T(x, y, z)$ are linear combinations of x, y, z.

Exercise 1.10

Fix a polynomial $q \in \mathcal{P}(\mathbf{R})$. Define $T : \mathcal{P}(\mathbf{R}) \to \mathcal{P}(\mathbf{R})$ by $(Tp)(x) = p(q(x))$ for each $x \in \mathbf{R}$. Show that T is a linear operator on $\mathcal{P}(\mathbf{R}$.

Proof:

Simply follows from the fact that polynomials are defined pointwise, so this is the proof that functions form a vector space (but with the point x replaced with the point $q(x)$ in $\mathbf{R}$).

2 Linear Map Lemma

Lemma 2.1

Let V be a finite-dimensional vector space and W a vector space. Suppose $\{v_1, ..., v_n\}$ is a basis of V and $w_1, ..., w_n$ any $n$vectors in W. Then there exists a unique linear map $T : V \to W$ such that

$$Tv_k = w_k$$

for each $k = 1, ..., n$.

The existence part of the Linear Map Lemma means that we can find a linear map that maps a basis to any vectors we wish. The uniqueness part says that a linear map is completely determined by its values on a basis.

Exercise 2.2

Prove Theorem 2.1 (*Hint:* Show that $Tv = \alpha_1 w_1 + ... + \alpha_n w_n$. Explain how to choose the scalars α_i, why this uniquely defines T, and show that T is linear.)

Proof:

Since $\{v_1, \ldots, v_n\}$ is a basis of V, every element $v \in V$ can be written uniquely as

$$v = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n$$

for some scalars $c_1, \ldots, c_n$.
We define a linear map $T : V \to W$ by setting

$$T(v) = c_1 w_1 + c_2 w_2 + \cdots + c_n w_n.$$

This is well-defined because the coefficients $c_1, \ldots, c_n$ are uniquely determined by the choice of basis in V.
To check linearity, let $v, v' \in V$ with

$$v = c_1 v_1 + \cdots + c_n v_n \quad \text{and} \quad v' = d_1 v_1 + \cdots + d_n v_n.$$

Then,

$$v + v' = (c_1 + d_1)v_1 + \cdots + (c_n + d_n)v_n.$$

Applying T, we get

$$T(v + v') = (c_1 + d_1)w_1 + \cdots + (c_n + d_n)w_n = T(v) + T(v').$$

Also, for any scalar λ,

$$T(\lambda v) = (\lambda c_1)w_1 + \cdots + (\lambda c_n)w_n = \lambda T(v).$$

Therefore, T is linear.
Finally, to see uniqueness, suppose there exists another linear map $S : V \to W$ such that $S(v_k) = w_k$ for each $k = 1, \ldots, n$. For any $v = c_1 v_1 + \cdots + c_n v_n$, linearity implies

$$S(v) = c_1 S(v_1) + \cdots + c_n S(v_n) = c_1 w_1 + \cdots + c_n w_n = T(v).$$

Hence, $S = T$, proving uniqueness.
This completes the proof.

Proposition 2.3

If $T : V \to W$ is a linear map, then $T(0) = 0$. (Hint: $T(0) = T(0 + 0)$)

Exercise 2.4

Prove the above.

Proof:

We have $T(0) = T(0 + 0) = T(0) + T(0)$, so subtracting $T(0)$ gives $T(0) = 0$, as desired.

Exercise 2.5

Fix $m, b \in \mathbf{R}$. Show that the function $f : \mathbf{R} \to \mathbf{R}$ defined by $f(x) = mx + b$ is a linear map if and only if $b = 0$. This shows that the linear functions of basic algebra are not the same as linear maps in the context of linear algebra.

Proof:

The only if is clear, since $f(0) = b$. Now, we show that when $b = 0$, $f(x) = mx$ is a linear function. But, this is obvious (do it yourself I don't want to write it out)...

Exercise 2.6

Let $M^{m\times n}(F)$ be the vector space of $m \times n$ matrices with entries in F. Given a matrix $A \in M^{m\times n}(F)$, we can define a map $T : M^{m\times 1}(F) \to M^{n\times 1}(F)$ by $T(x) = Ax$ by matrix multiplication. Show that this is linear (*Hint:* Write this in components: the $T(x)_i = \sum_{k=1}^{n} A_{ij}x_j$.).

Proof:

Each component is a linear combination the x_j, and therefore, by linearity, we see:

$$\begin{aligned} T(\alpha x + y)_i &= \sum_{j=1}^{n} A_{ij}(\alpha x_j + y_j) \\ &= \sum_{j=1}^{n} A_{ij}\alpha x_j + \sum_{j=1}^{n} A_{ij}y_j \\ &= \alpha \sum_{j=1}^{n} A_{ij}x_j + \sum_{j=1}^{n} A_{ij}y_j \\ &= \alpha T(x)_i + T(y)_i, \end{aligned}$$

and extending this in each component gives the desired result.

3 Algebra of Linear Maps

Fix vector spaces V and W over F. Let $\mathcal{L}(V, W)$ denote the set of all linear maps $T : V \to W$. In the case $W = V$, we write $\mathcal{L}(V) := \mathcal{L}(V, V)$. Given $S, T \in \mathcal{L}(V, W)$ and $\alpha \in F$, we define the sum $S + T$ and scalar multiple αT by:

$$\begin{aligned} (S + T)(v) &:= Sv + Tv \\ (\alpha T)(v) &:= \alpha Tv. \end{aligned}$$

Another common notation for $\mathcal{V}, \mathcal{W}$ which you may encounter elsewhere is HomV, W). This is a more general notation which comes from category theory, emphasizing that linear maps are the morphisms in the category of vector spaces over F.

Exercise 3.1

Prove that $\mathcal{L}(V, W)$ is a subspace of W^V, the vector space of all functions $V \to W$. Note that we showed last time that the zero map is linear, so $0 \in \mathcal{L}(V, W)$. All that remains to show is that $\mathcal{L}(V, W)$ is closed under the operations above (or, equivalently, under linear combinations).

Proof:

Simply, if f, g are linear, then $\alpha f + g$ are linear. But, we know already that:

$$(\alpha f + g)(x) = \alpha f(x) + g(x),$$

so we have:

$$\begin{aligned}(\alpha f+g)(\beta x+y) &= \alpha f(\beta x+y)+g(\beta x+y)\\ &= \alpha(\beta f(x)+f(y))+\beta g(x)+g(y)\\ &= \beta(\alpha f)(x)+(\alpha f)(y)+\beta g(x)+g(y)\\ &= \beta((\alpha f)(x)+g(x))+(\alpha f)(y)=g(y)\\ &= \beta((\alpha f+g)(x)+(\alpha f+g)(y),\end{aligned}$$

showing the subspace criterion/



It usually makes no sense to multiply two elements of a vector space, but for some pairs of linear maps a useful product does exist.



Definition 3.2 (Product of Linear Maps)

. If $T \in \mathcal{L}(U,V)$ and $S \in \mathcal{L}(V,W)$, then we can define the *product* $ST \in \mathcal{L}(U,W)$ by $ST := S \circ T$. That is, for each $u \in U$,

$$(ST)(u) = S(T(u)).$$

Note that, just as for ordinary functions, ST is only defined when the range of T is contained in the domain of S.



Exercise 3.3

If $T \in \mathcal{L}(U,V)$ and $S \in \mathcal{L}(V,W)$, prove that $ST \in \mathcal{L}(U,W)$. That is, prove that ST is linear.



Proof:

We have:

$$ST(\alpha u+v) = S(T(\alpha u+v)) = S(\alpha Tu+Tv) = \alpha S(Tu)+S(Tv) = \alpha ST(u)+ST(v),$$

showing linearity.





Exercise 3.4

Let $D \in \mathcal{L}(\mathcal{P}(\mathbf{R}))$ be differentiation and $T \in \mathcal{L}(\mathcal{P}(\mathbf{R}))$ be multiplication by x^2. We showed last time that both of these are linear maps. Show that $TD \neq DT$. This shows that the product of linear maps, in general, is not commutative.



Solution 2

We have that $TD(p(x)) = T(p'(x)) = x^2p'(x)$. But, we have $DT(p(x)) = D(x^2p(x)) = 2xp(x) + x^2p'(x)$, and $2xp(x) \neq x^2p'(x)$ in $\mathbf{R}$ (it does over a field of char 2, but that is a different story...).

In the case of $\mathcal{L}(V)$, the product of two linear operators is always defined. It is straightforward to verify that the product of linear operators on V has the following properties:

(a) (Bilinearity) For all $S, T_1, T_2 \in \mathcal{L}(V)$ and all $\alpha \in F$,

- $S(T_1+T_2) = ST_1 + ST_2$

- $(S_1 + S_2)T = S_1T + S_2T$
- $(\alpha S)T = \alpha(ST) = S(\alpha T)$

(b) (Associativity) $(RS)T = R(ST)$ for all $R, S, T \in \mathcal{L}(V)$.

(c) (Identity) $IT = TI = T$ for all $T \in \mathcal{L}(V)$, where I is the identity map.

These properties are said to make $\mathcal{L}(V)$ into a *unital associative* F-algebra.